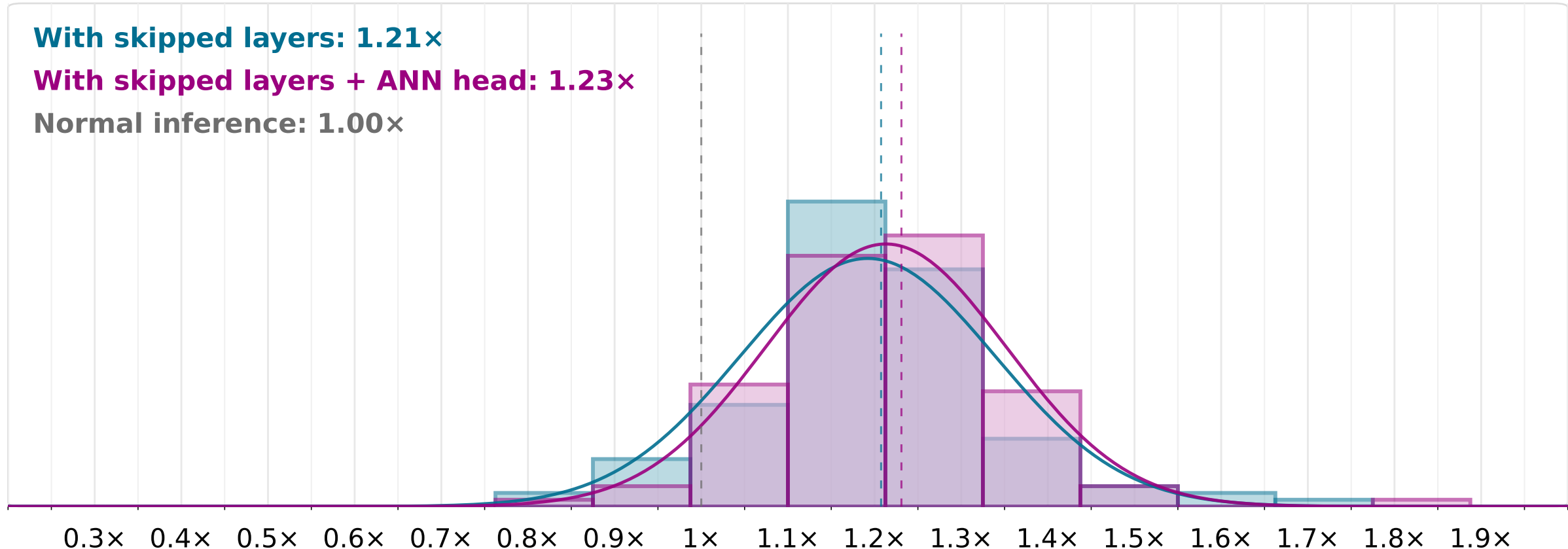


Self-speculation per-token speedup

Llama 3.2 1B Instruct



Setup

Prompt set = **concrete**
Gap = **(1, 1)**
Block size = **2**
Layers = **16**
LM vocab = **128,256**
LM params = **1.24B**
Head params = **262.7M**
Head portion = **21.25%**

Runtime

GPU = **NVIDIA L40S**
Backend = **sdpa**
Model dtype = **float32**
Bridge dtype = **float32**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **120**

Results

Peak mem normal = **4.66 GiB**
Mean speedup (w/o ANNH) = **1.21×**
Acceptance rate (w/o ANNH) = **45.0%**
Exact match (w/o ANNH) = **100.0%**
Peak mem self (w/o ANNH) = **4.67 GiB**
Mean speedup (w ANNH) = **1.23×**
Acceptance rate (w ANNH) = **42.1%**
Exact match (w ANNH) = **100.0%**
Peak mem self (w ANNH) = **4.74 GiB**
Tokens gen = **s 16,811 / n 16,811**

Profile

Drafter split (w/o ANNH) = **B 52.6% / H 38.6% / O 8.8%**
Verifier/normal (w/o ANNH) = **1.05×**
Drafter/normal (w/o ANNH) = **34.2%**
Drafter split (w ANNH) = **B 76.6% / H 11.2% / O 12.2%**
Verifier/normal (w ANNH) = **1.04×**
Drafter/normal (w ANNH) = **23.2%**
ANNH accuracy = **93.6%**
ANNH head speedup = **4.81×**
ANNH index = **8016 clusters; top-k 70**